STA 2002

Lecture 10

Test of proportion

Test of a single proportion

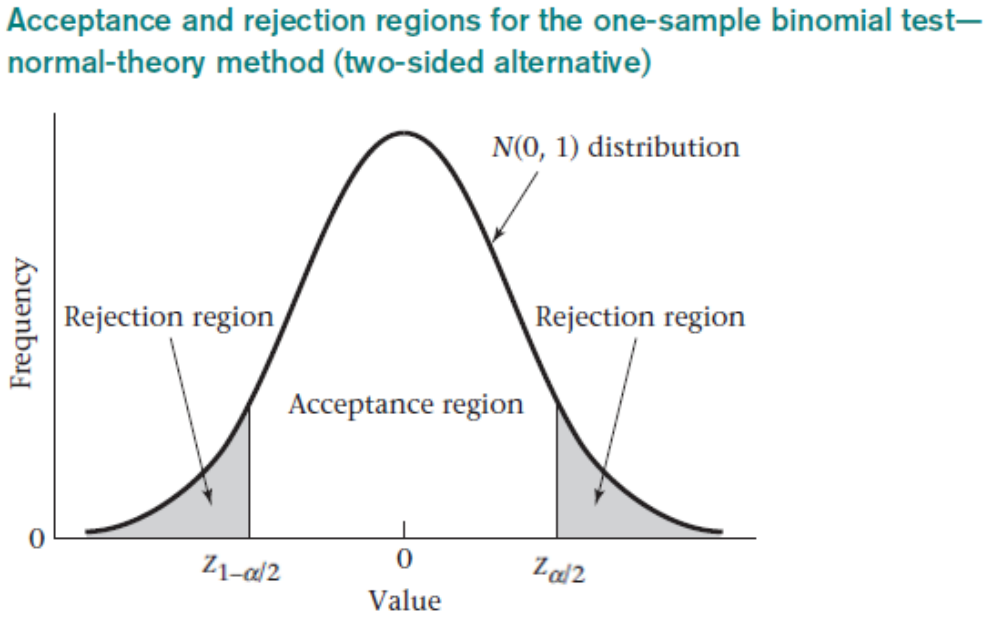
$H_0: p = p_0$
$H_1: p \neq p_0$

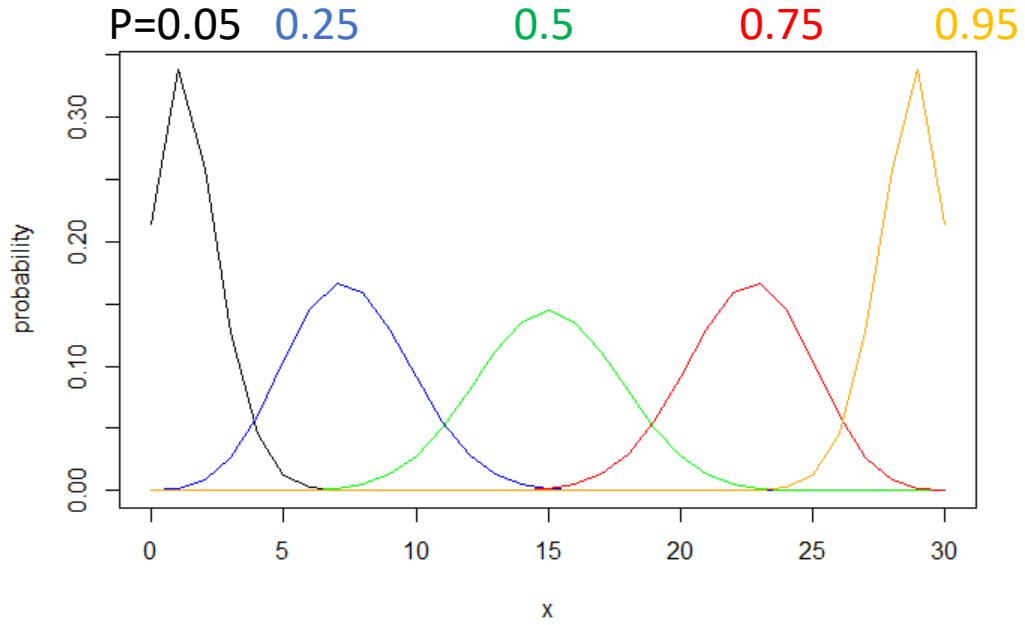
relevant statistic	$\hat{\boldsymbol{p}} = \frac{x}{n}$
Distribution of $\hat{\boldsymbol{p}}$ under the null	Approximately normal, $N(p_0, \frac{p_0(1-p_0)}{n})$
Distribution of test statistic under the null	$\frac{\hat{\boldsymbol{p}} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0,1)$

Reminder:

X is binomial (n, p), then

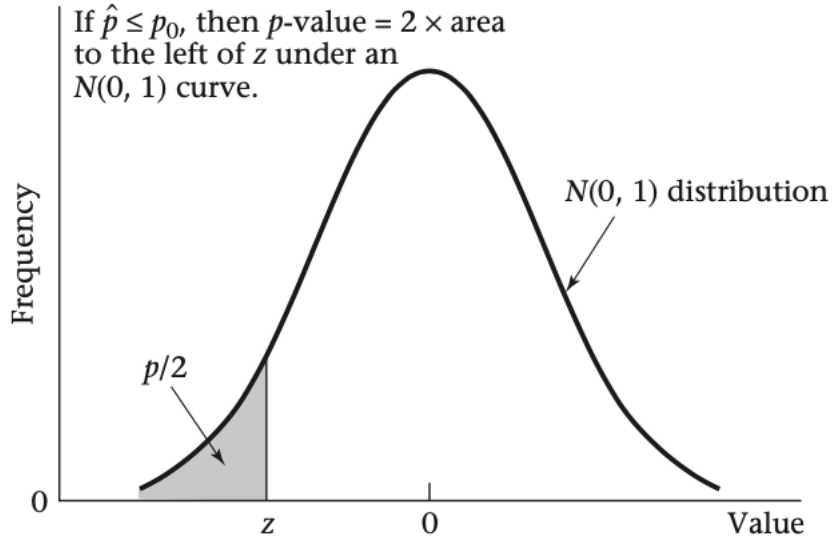
E(X)=np, var(X)=np(1-p)

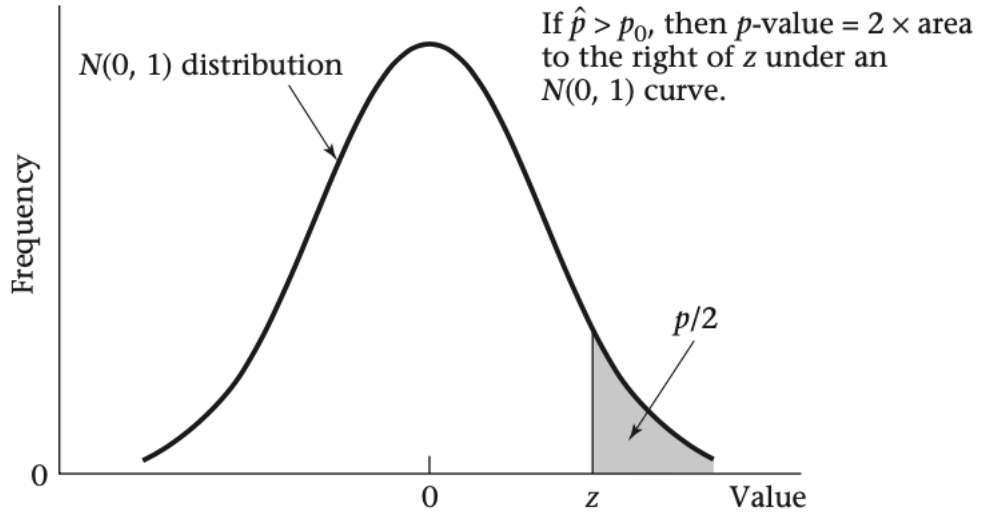




Test of a single proportion

Illustration of the *p*-value for a one-sample binomial test— normal-theory method (two-sided alternative)





Test of a single proportion

It is suspected that some genetic factor contributes to breast cancer risk. Suppose that 400 of the 10,000 women ages 50–54 whose mothers had breast cancer had breast cancer themselves at some time in their lives. Given large studies, assume the prevalence rate of breast cancer for U.S. women in this age group is about 2%. The question is: How compatible is the sample rate of 4% with a population rate of 2%?

$$H_0: p = .02$$
$$H_1: p \neq .02$$

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{.04 - .02}{\sqrt{\frac{.02(1-.02)}{10000}}} = \frac{.02}{.0014} = 14.3$$

Because $Z_{0.025} = 1.96$ and 14.3>1.96, it follows that H_0 can be rejected using a two-sided test with $\alpha = 0.05$.

$$p\text{-value} = 2 \times [1 - \Phi(z)] = 2 \times [1 - pnorm(14.3)] \approx 0$$

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> 2*(1-pnorm(14.3))
[1] 0
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In the above, $\Phi(x) = P(Z \leq x)$ is the cdf for the standard normal distribution.

Test for difference in proportion

Distribution of $\widehat{p_1} = \frac{x_1}{n_1}$:	**~Normal,** $N(p_1, \frac{p_1(1-p_1)}{n_1})$

Distribution of $\widehat{p_2} = \frac{x_2}{n_2}$	**~Normal,** $N(p_2, \frac{p_2(1-p_2)}{n_2})$

Distribution of $\widehat{p_1} - \widehat{p_2} = \frac{x_1}{n_1} - \frac{x_2}{n_2}$	**~Normal,** $N(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2})$
Distribution of $\widehat{p_1} - \widehat{p_2}$ under the null H_0: $p_1 = p_2$	$\frac{\widehat{p_1} - \widehat{p_2}}{\sqrt{\widehat{\boldsymbol{p}}(1-\widehat{\boldsymbol{p}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$

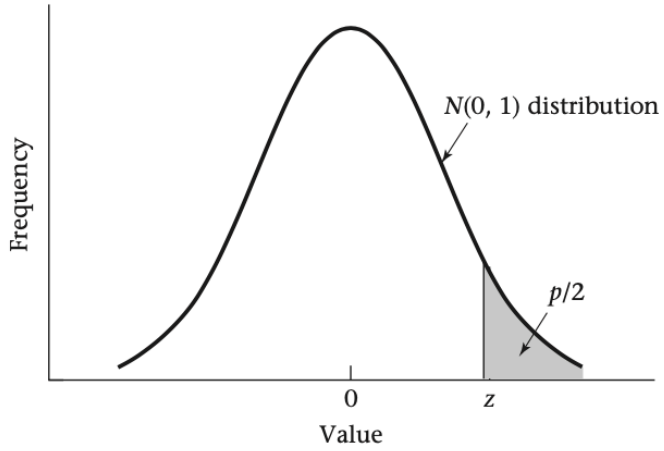
$$\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2}$$

Under the null, we assume $p_1 = p_2 = p_0$, but we don't know p_0.
So we use $\widehat{\boldsymbol{p}}$ to plug in for p_0.

Test for difference in proportion

Cardiovascular Disease A study looked at the effects of OC use on heart disease in women 40 to 44 years of age. The researchers found that among 5000 current OC users at baseline, 13 women developed a myocardial infarction (MI) over a 3-year period, whereas among 10,000 non-OC users, 7 developed an MI over a 3-year period. Assess the statistical significance of the results.

Computation of the exact *p*-value for the two-sample test for binomial proportions (normal-theory test)



$H_0: p_1 = p_2$ vs. $H_1: p_1 \neq p_2$

$n_1 = 5000,\ \hat{p}_1 = 13/5000 = .0026,\ n_2 = 10{,}000,\ \hat{p}_2 = 7/10{,}000 = .0007.$

$$\hat{p} = \frac{13+7}{15{,}000} = \frac{20}{15{,}000} = .00133$$

$$z = \frac{.0026 - .0007}{\sqrt{.00133(.99867)(1/5000 + 1/10{,}000)}} = \frac{.00175}{.00063} = 2.77$$

The p-value is given by $2 \times [1 - \Phi(2.77)] = 2 \times [1 - pnorm(2.77)] = .006$

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> 2*(1-pnorm(2.77))
[1] 0.005605629
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